

| PAPER CODE | EXAMINER | DEPARTMENT | TEL |
|------------|----------|------------|-----|
| INT202     |          |            |     |

**2nd SEMESTER 2021/22 FINAL EXAMINATION**

**UNDERGRADUATE – YEAR 3**

**COMPLEXITY OF ALGORITHMS**

**TIME ALLOWED: 2 HOURS**

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**INSTRUCTIONS TO CANDIDATES**

- 1. This is an open-book examination.**
- 2. The examination will be conducted in all other respects in accordance with the Regulations for the Conduct of Examinations.**
- 3. Total marks available are 100. This exam will count for 80% in the final assessment.**
- 4. Answer all questions. There is NO penalty for providing a wrong answer.**
- 5. Only English solutions are accepted.**

**Question 1. [10 MARKS]**

What do the following algorithms Foo and Bar do? Analyze their worst-case running times respectively, and express them using "Big-Oh" notation.  
[10 marks]

```
Algorithm Foo (a, n):
  Input: two integers, a and n
  Output: ?
  k ← 0
  b ← 1
  while k < n do
    k ← k + 1
    b ← b * a
  return b
```

```
Algorithm Bar (a, n):
  Input: two integers, a and n
  Output: ?
  k ← n
  b ← 1
  c ← a
  while k > 0 do
    if k mod 2 = 0 then
      k ← k/2
      c ← c * c
    else
      k ← k - 1
      b ← b * c
  return b
```

**Question 2. [5+5+5=15 MARKS]**

Give a tight asymptotic upper bound ( $\Theta$  notation) on the solution to each of the following functions. You need to justify your answers.

(1) [5 marks]

$$T1(\text{int } n) = \begin{cases} 1, & \text{if } n = 1 \\ T1(n-1) + n(n-1), & \text{if } n \geq 2 \end{cases}$$

(2) [5 marks]

```
T2(int n){
  if(n==1) return 1;
  for (int i=1;i<=n;i++)
    for (int j=1;j<=n;j++)
      print("*");
  function T2(n-3);
}
```

(3) [5 marks]

```

T3 (int n){
    if(n<=2) return 1;
    else return (T3(floor(sqrt(n)))+1);
}

```

**Question 3. [3+3=6 MARKS]**

Let  $T$  be a  $k$ -ary tree, i.e. each internal node of  $T$  has at most  $k$  children. Suppose that the height of  $T$  is  $h$ .

a. What is the maximum number of external nodes (leaves) that  $T$  can have? Explain your answer.

[3 marks]

b. Compute the maximum total number of nodes that  $T$  can have.

[3 marks]

**Question 4. [8+7=15 MARKS]**

(1) Given the following integer elements:

8, 1, 0, 6, 9, 7, 4, 2, 5, 3

Draw the AVL tree that results when the above elements are added (in the given order) to an empty AVL tree (intermediate steps are optional). What is the running time (in Big-Oh notation) of an insertion operation in an AVL tree?

[8 marks]

(2) Draw the final AVL tree from (1) after all of the elements  $\{3, 6, 8, 1\}$  are removed (intermediate steps are optional). What is the running time (in Big-Oh notation) of a removal operation in an AVL tree?

[7 marks]

**Question 5. [4+10=14 MARKS]**

Let set  $S = \{S_k\}$ ,  $k=1, 2, 3, 4$ , denote a set of objects with weights and benefits as given in the table below.

|       |    |    |    |    |
|-------|----|----|----|----|
| $k$   | 1  | 2  | 3  | 4  |
| $b_k$ | 25 | 15 | 20 | 36 |
| $w_k$ | 7  | 2  | 3  | 6  |

For each of the following questions, write down the answers with a brief analysis.

(1) What is an optimal solution to the fractional Knapsack problem for  $S$  assuming that we have a sack that can hold objects with total weight 10?

[4 marks]

(2) What is an optimal solution to the 0-1 Knapsack problem for  $S$  assuming that we have a sack that can hold objects with total weight 10. Solve the problem with dynamic programming.

[10 marks]

### Question 6. [10 MARKS]

For a given flow network, the Ford-Fulkerson algorithm enables us to find the maximum flow over the network. What is the worst-case running time of the Ford-Fulkerson algorithm if all edge capacities are bounded by a constant  $C$ ?

[10 marks]

### Question 7. [2+5+8+5=20 MARKS]

Alice and Bob wish to use the RSA cryptosystem for digital signatures.

(1) Compute  $n$  and  $\Phi(n)$  associated with the prime numbers  $p = 17$  and  $q = 23$ .

[2 marks]

(2) Among the values  $e = 11$  and  $e = 13$ , which one can be used as a public key? Compute the associated private key  $d$ . Please justify your answer.

[5 marks]

(3) Assume Alice's public key is the value  $e$  obtained in (2). Alice wants to sign the message  $M=100$ . What message should she send to Bob? Please justify your answer.

[8 marks]

(4) How does Bob perform the verification? Please justify your answer.

[5 marks]

### Question 8. [10 MARKS]

Consider the following problem: Let  $G$  be a directed graph. Let  $S$  be a subset of vertices of  $G$  such that deletion of  $S$  (vertices) and the edges incident to vertices in  $S$  results in a graph  $G'$  with no directed cycles. Such an  $S$  is a feedback node set. The size of  $S$  is the number of vertices in  $S$ . The Feedback Node Set (FNS) decision problem is "to determine for a given integer  $k$  if  $G$  has a feedback node set of size at most  $k$ ". Show how VC can be reduced to FNS in polynomial time. Discuss the complexity of the FNS decision problem.

[10 marks]

**END OF EXAM PAPER**